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(54) **INSTRUMENT RELEASE**

(71) Applicant: **Intuitive Surgical Operations, Inc.**,
Sunnyvale, CA (US)

(72) Inventors: **Bram Gilbert Antoon Lambrecht**,
Redwood City, CA (US); **Robert E. Holop**, Santa Clara, CA (US)

(73) Assignee: **Intuitive Surgical Operations, Inc.**,
Sunnyvale, CA (US)

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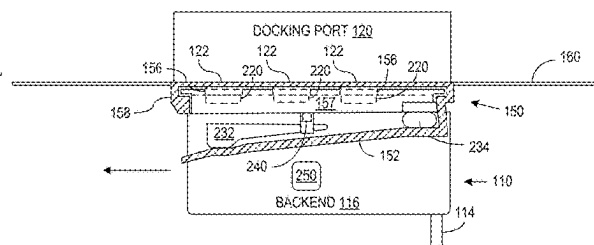
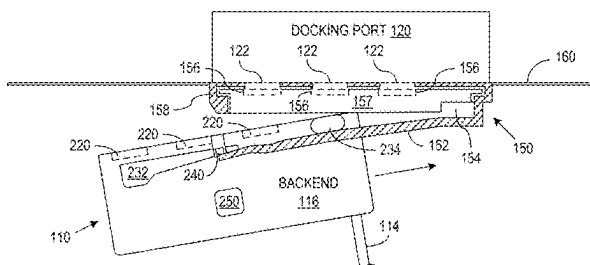
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Primary Examiner — Daniel J Wiley

(57) **ABSTRACT**

A medical instrument includes a backend and a pair of
release elements. The backend includes a docking feature
shaped to engage a docking structure such as a docking port
of a surgical robot or a sterile adaptor for a surgical robot.
Each release element is coupled to the backend and may
include an activation feature on one side of the backend and
a push tab extending from either an opposite side or the same
side of the backend. Movement of the activation features
moves the push tabs from being aligned with the docking
feature to pushing on the docking structure, permitting the
medical instrument to be disengaged from the docking
structure. Optionally, a coupling of the release elements
limits a movement of the activation features.

20 Claims, 10 Drawing Sheets



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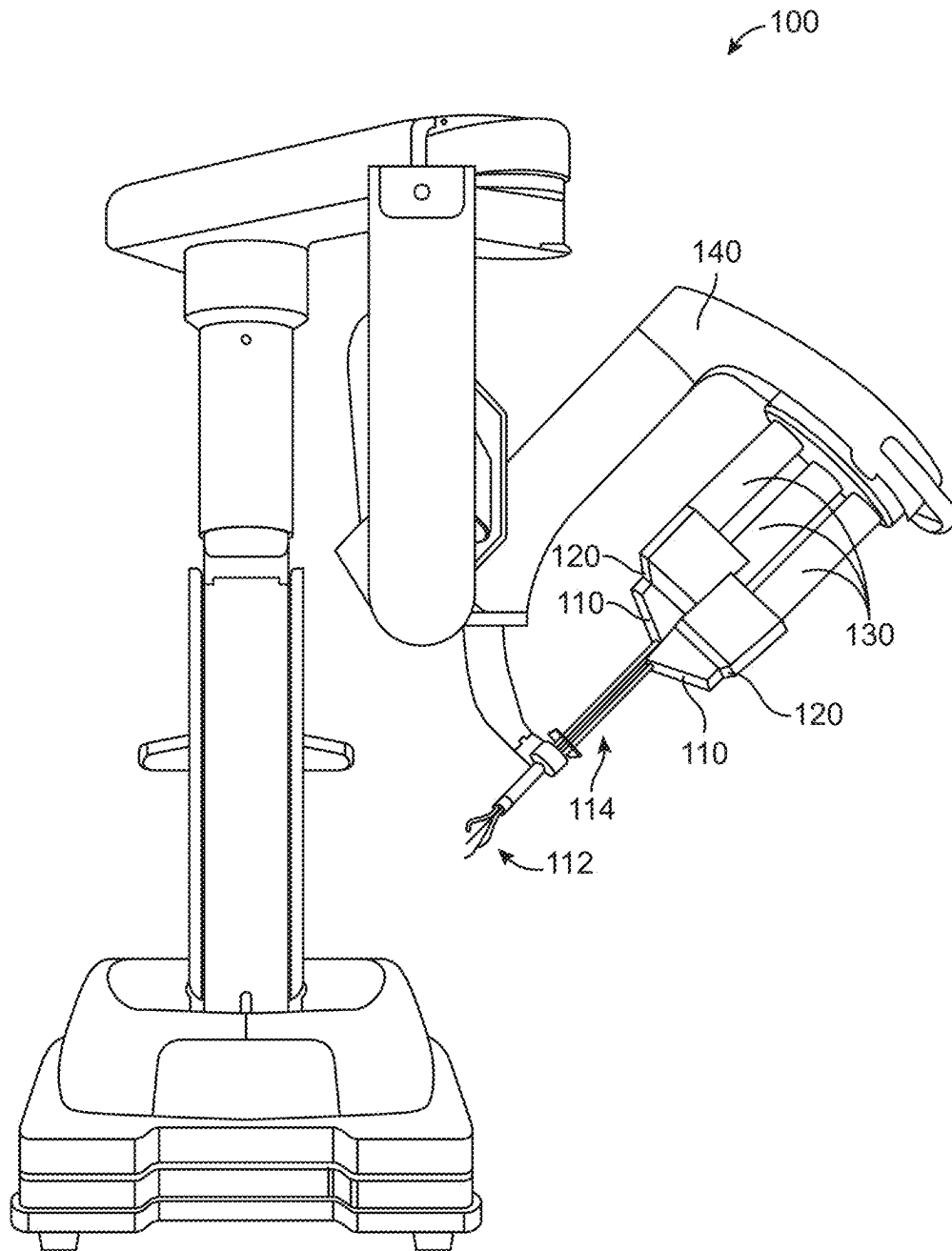
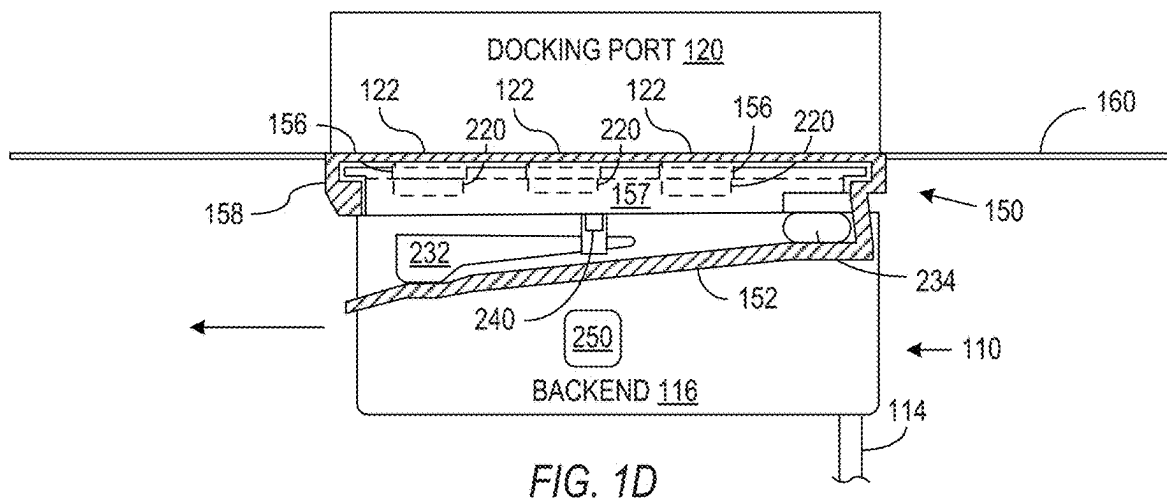
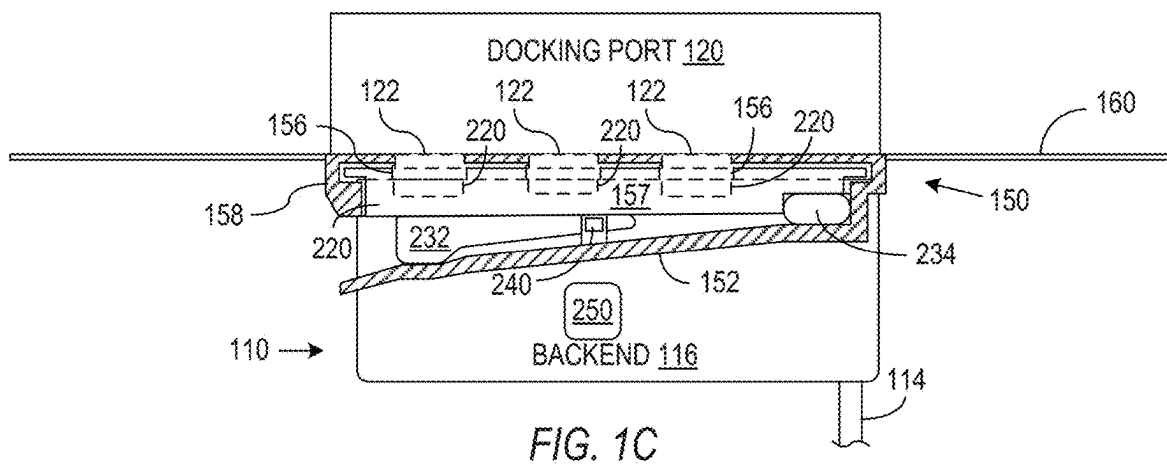
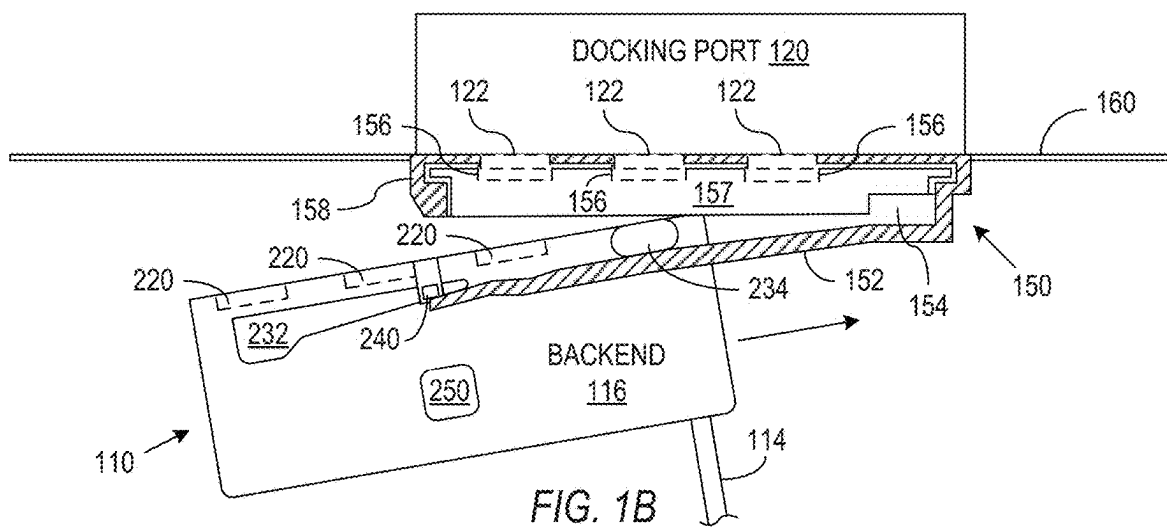
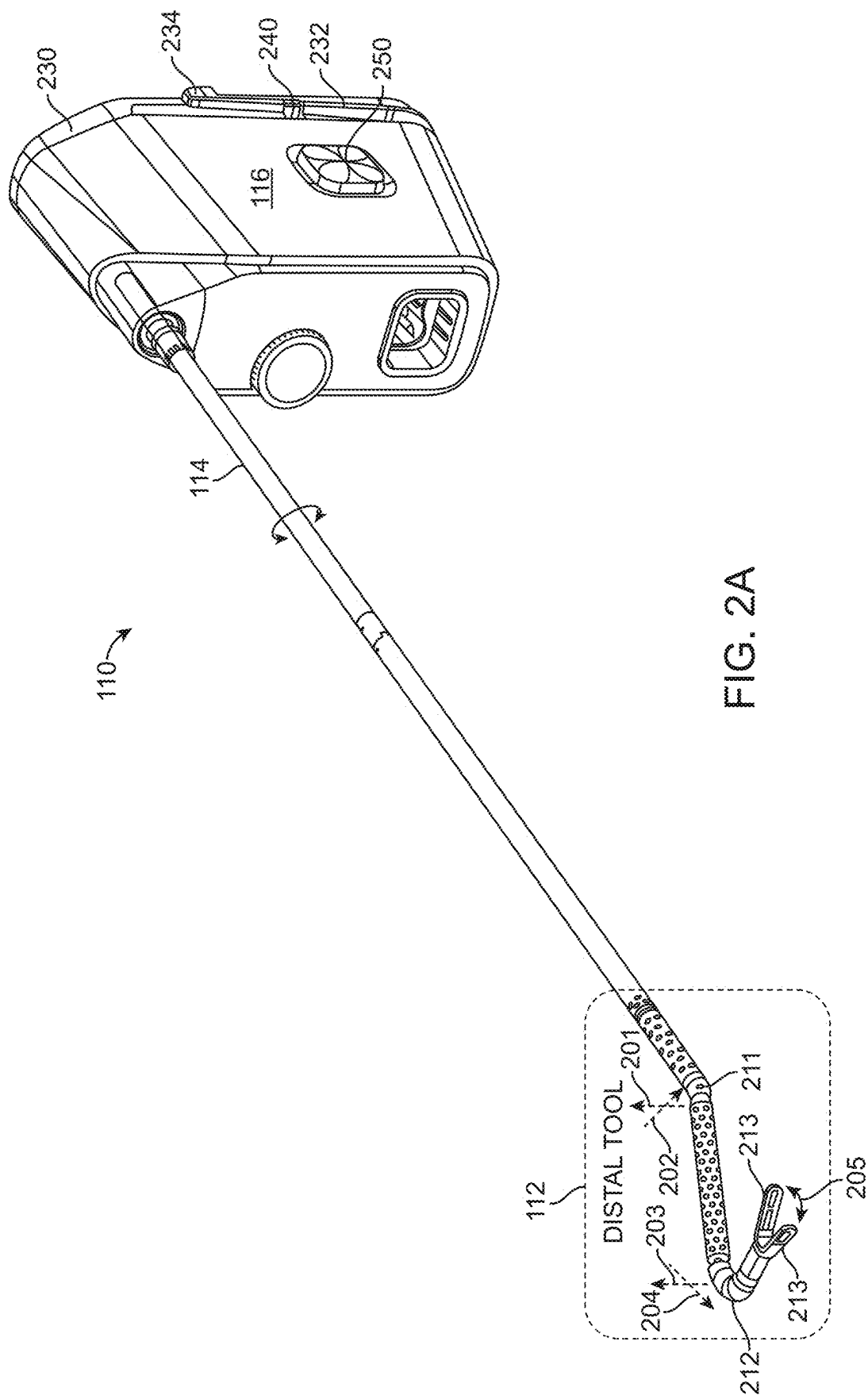


FIG. 1A





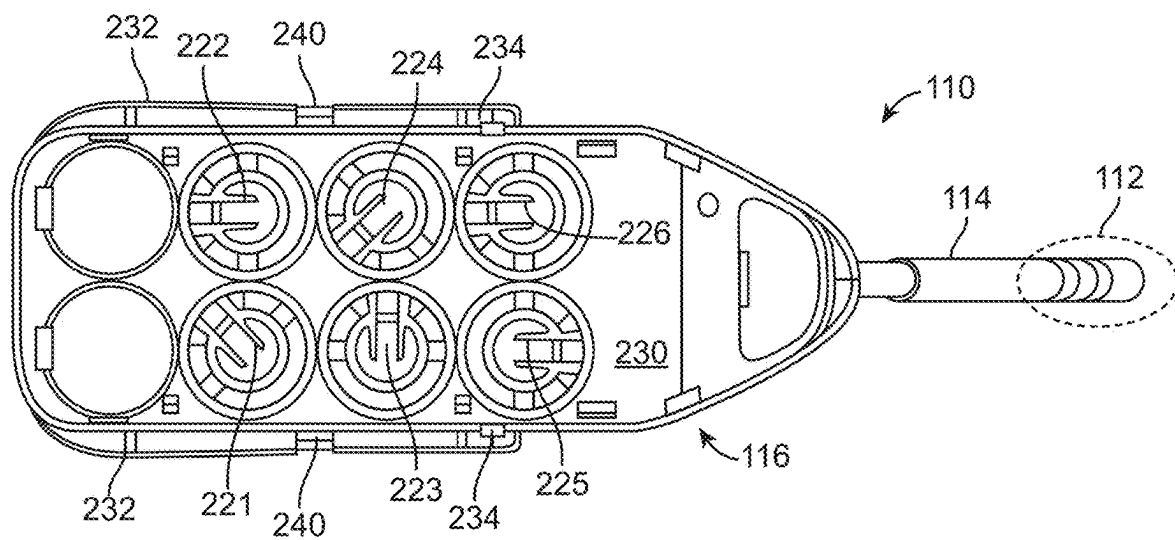


FIG. 2B

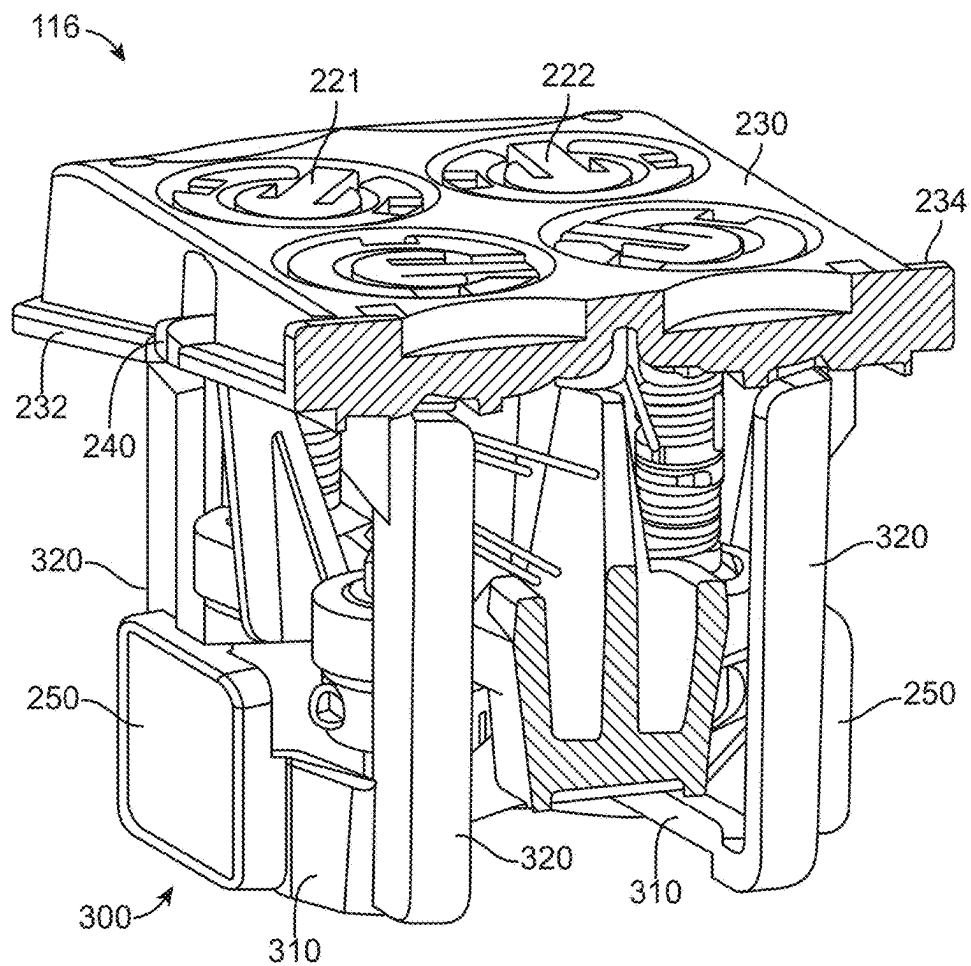


FIG. 3A

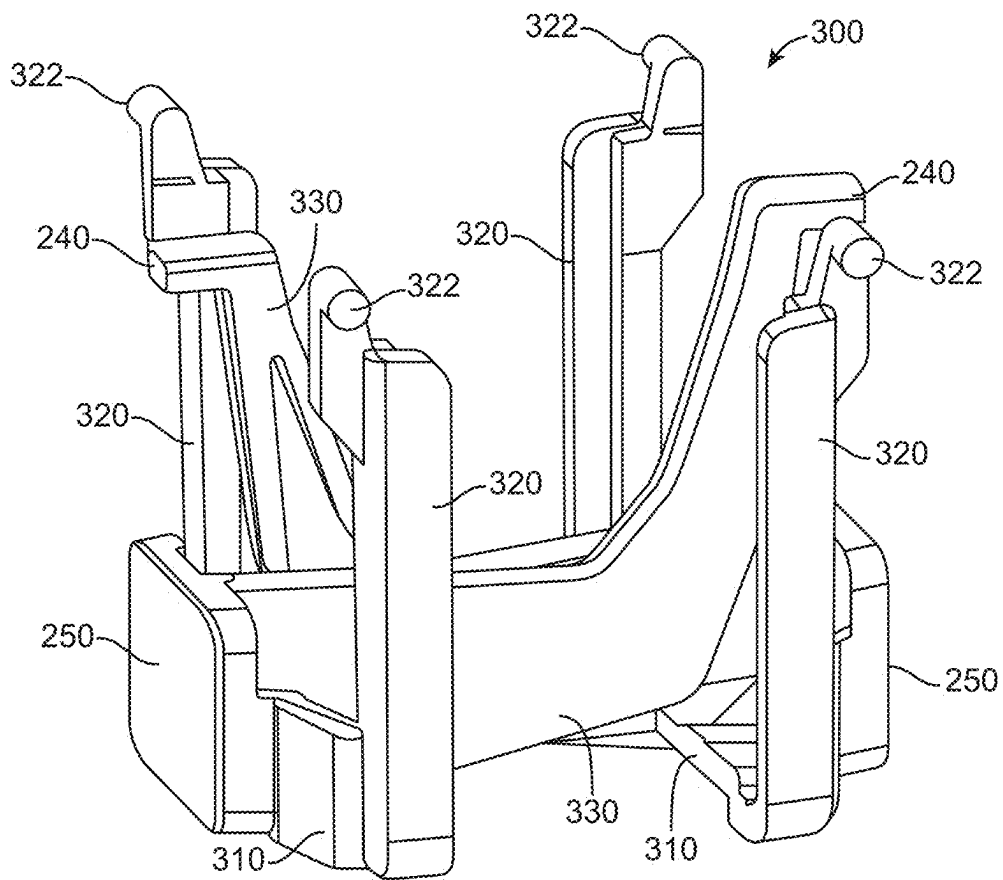


FIG. 3B

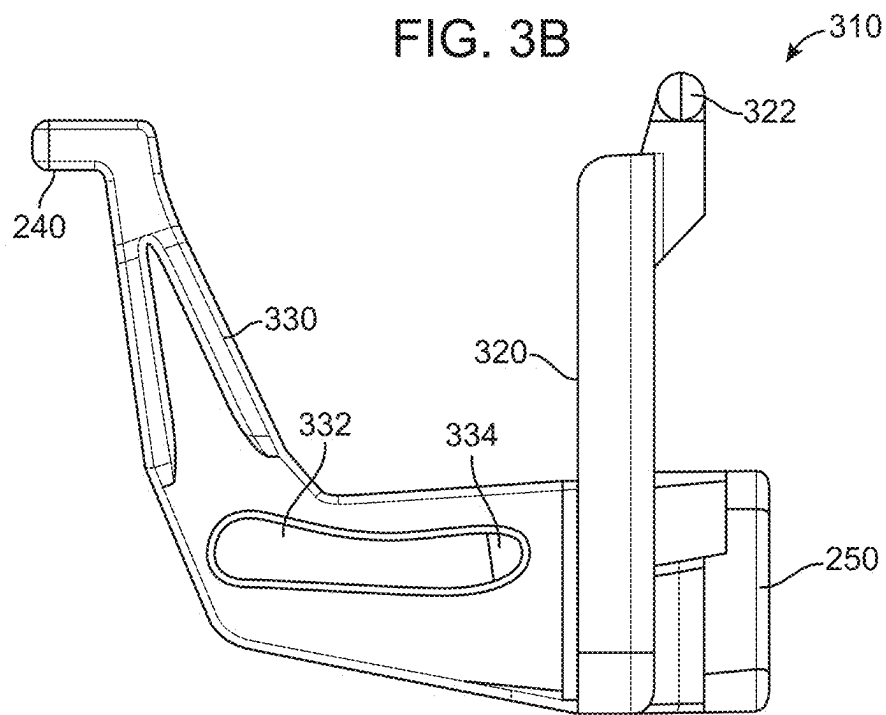


FIG. 4A

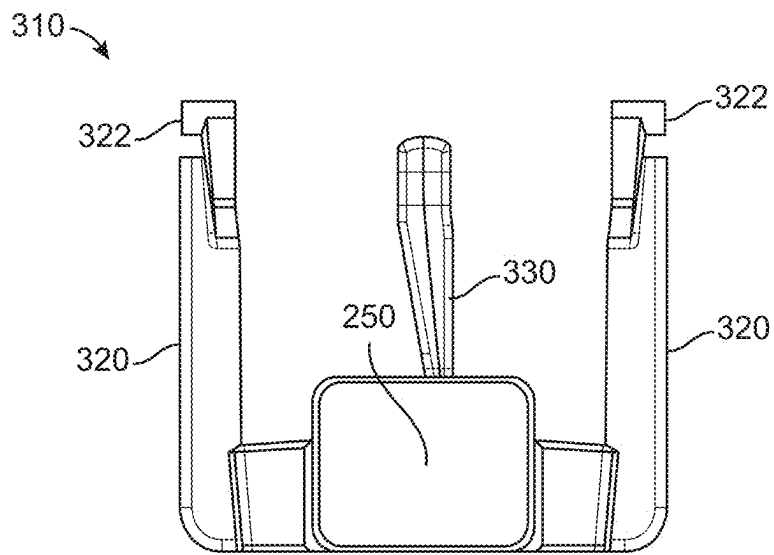


FIG. 4B

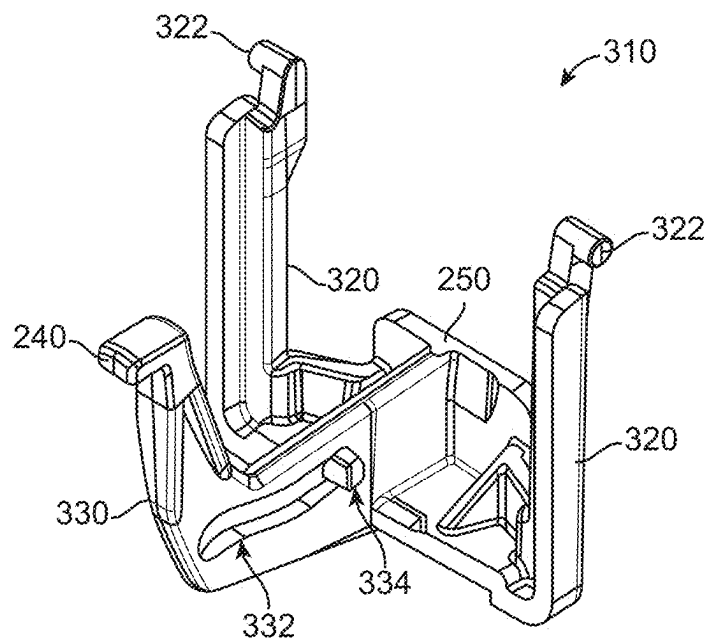


FIG. 4C

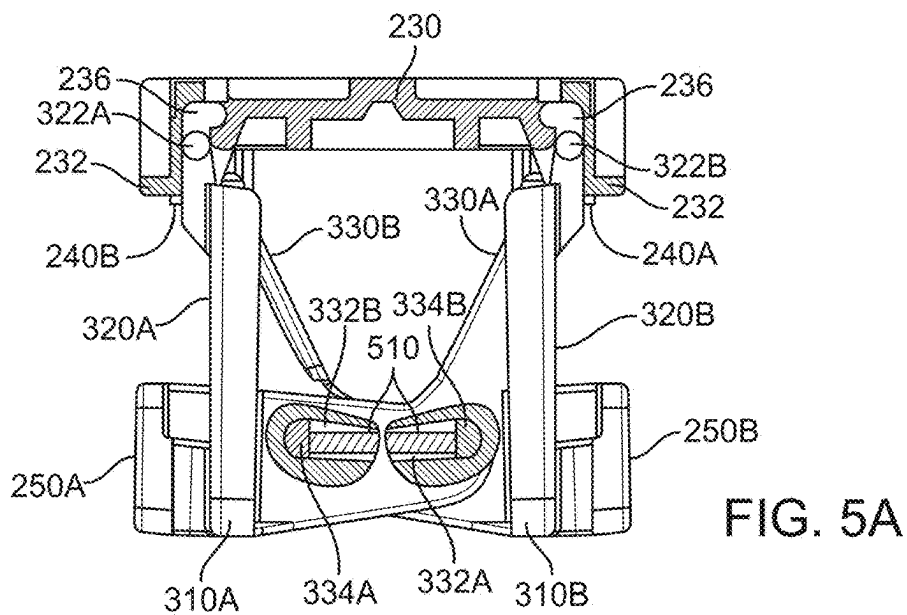


FIG. 5A

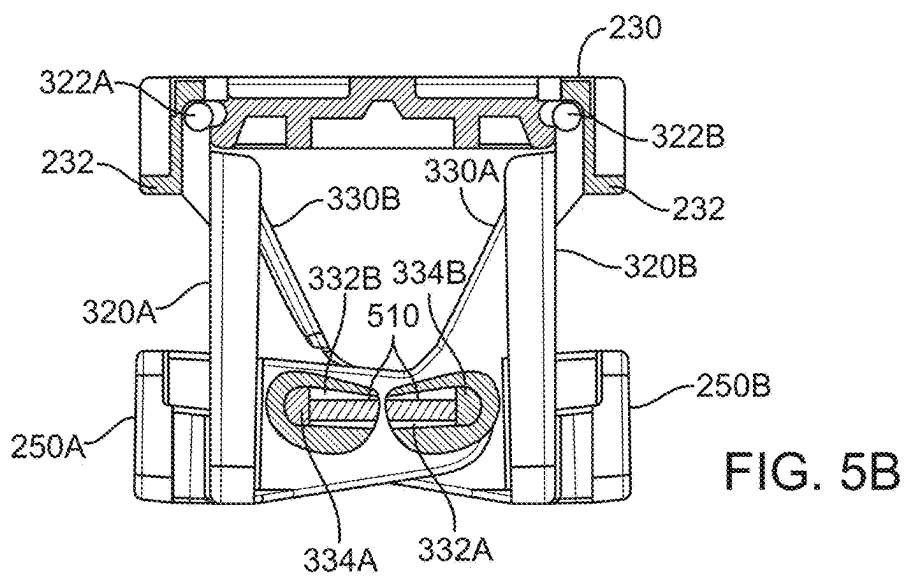


FIG. 5B

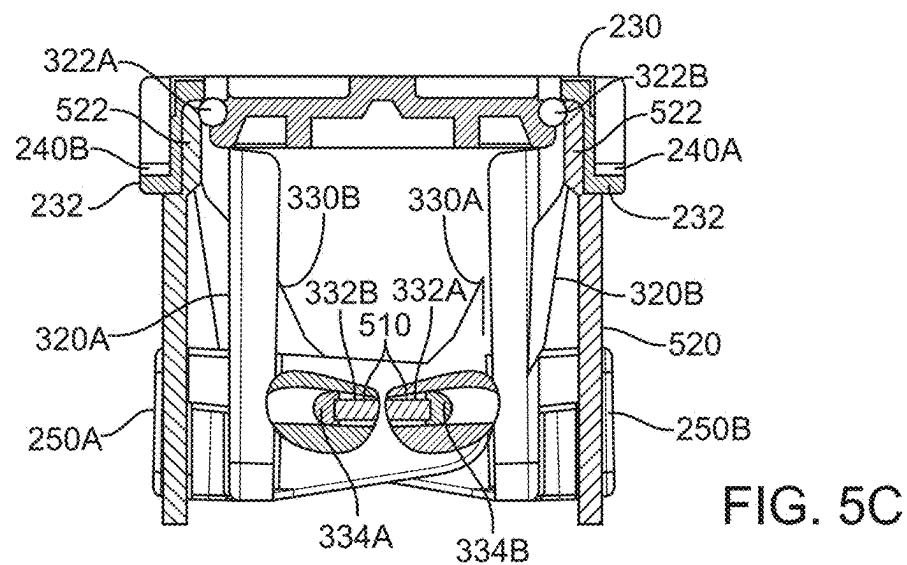


FIG. 5C

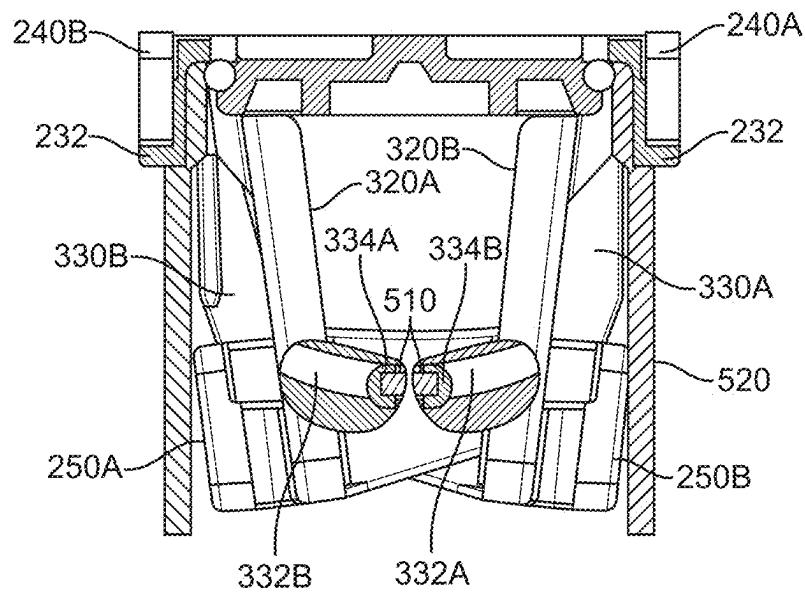


FIG. 5D

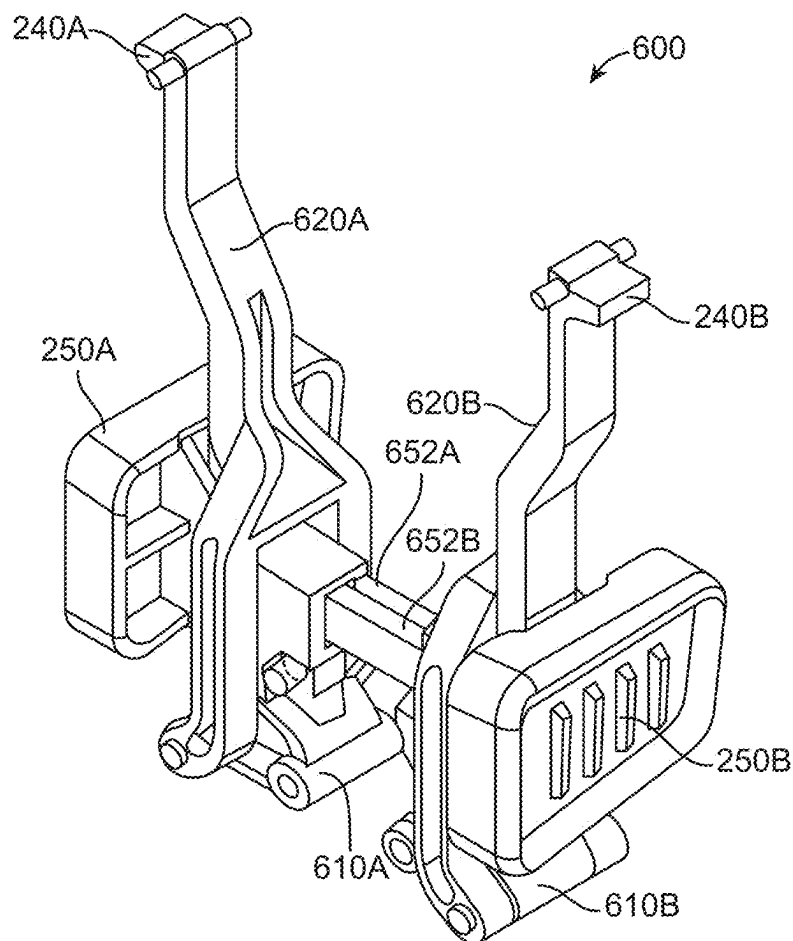


FIG. 6A

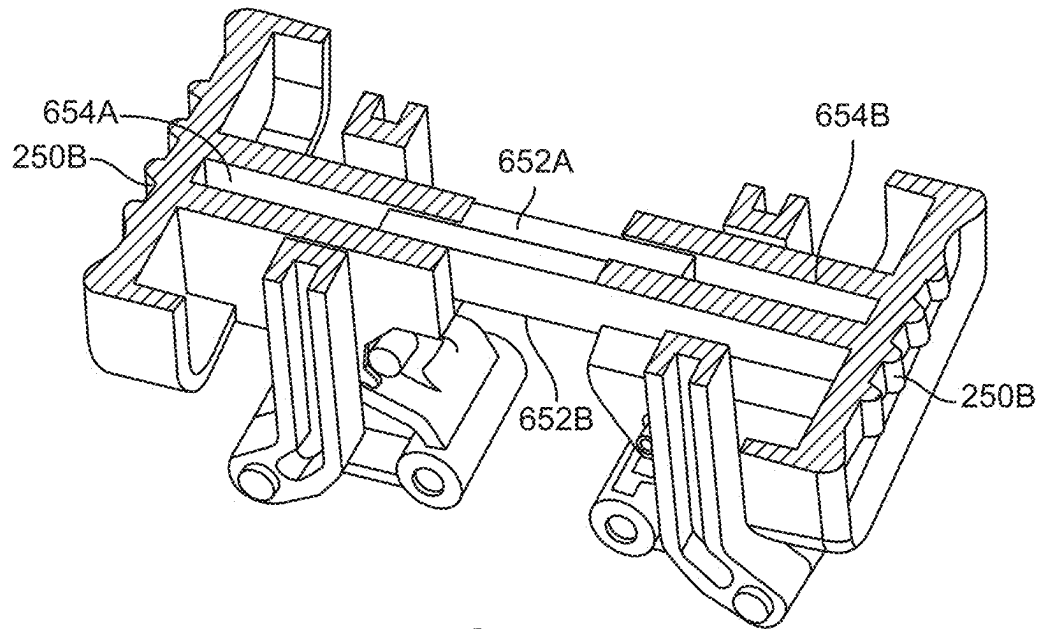


FIG. 6B

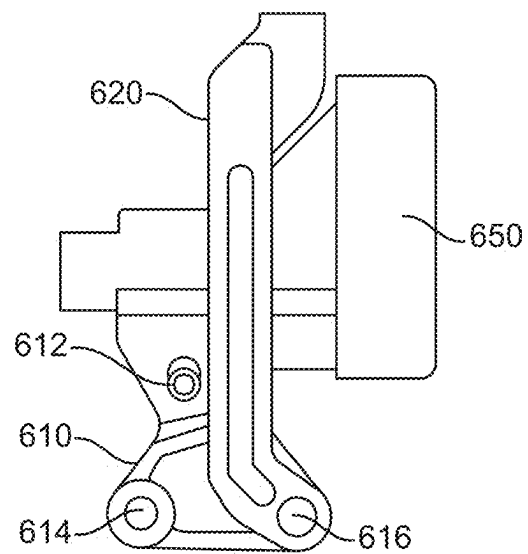


FIG. 6C

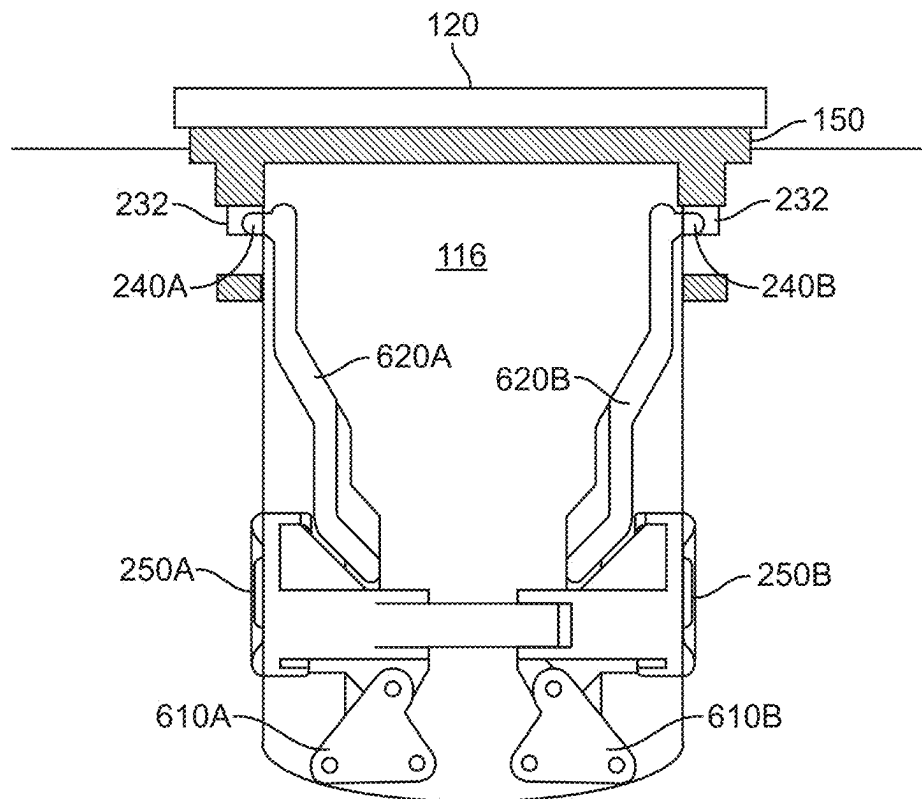


FIG. 6D

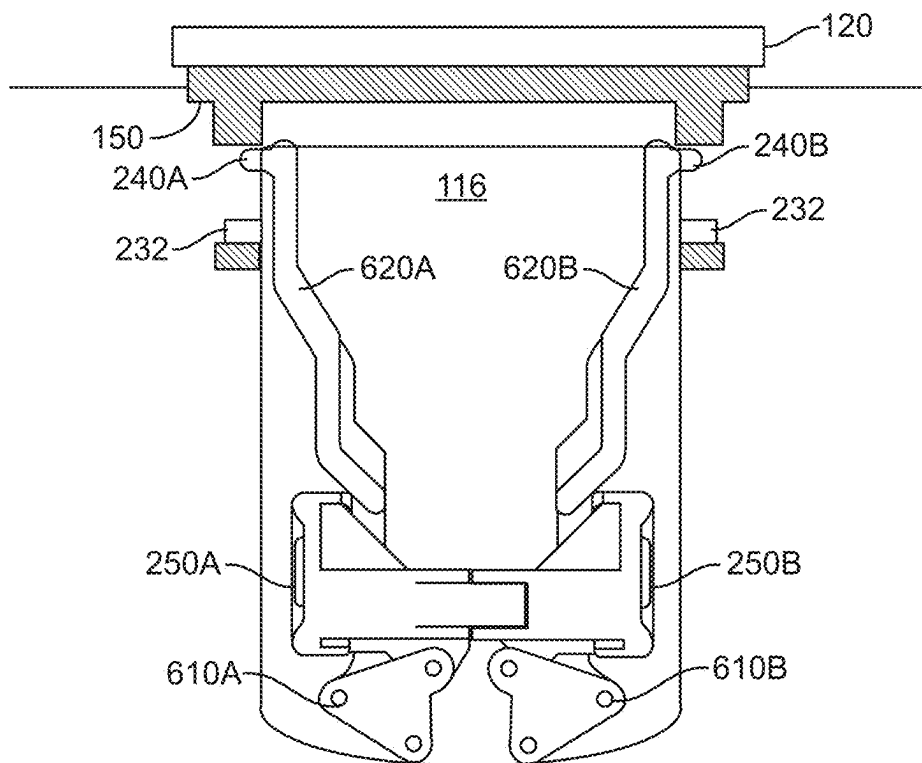


FIG. 6E

INSTRUMENT RELEASE

CROSS-REFERENCE TO RELATED APPLICATIONS

This patent application is a continuation of U.S. application Ser. No. 16/317,243 (filed Jan. 11, 2019), entitled “INSTRUMENT RELEASE,” which is a U.S. national stage filing under 35 U.S.C. § 371 of International Application No. PCT/US2017/024687 (filed Mar. 29, 2017), entitled “INSTRUMENT RELEASE,” which claims priority to and the benefit of the filing date of U.S. Provisional Patent Application No. 62/362,454 (filed Jul. 14, 2016), entitled “INSTRUMENT RELEASE,” each of which is incorporated by reference herein in its entirety.

BACKGROUND

Some advanced robotic medical systems such as the da Vinci® Surgical System commercialized by Intuitive Surgical, Inc. employ removable medical instruments that may be mounted in docking ports on the arms of a robot. (As used herein, the terms “robot” or “robotically” and the like include teleoperation or telerobotic aspects, and teleoperated surgical systems are often referred to as “surgical robots”.) Such systems generally employ a sterile barrier between the medical instruments and the robot, so that medical instruments, which come into direct contact with a patient, reside in a sterile environment on a patient side of the sterile barrier, while the robot, which never directly contacts a patient, is outside of the sterile environment. The sterile barrier typically includes a drape or sheet of plastic or other material suitable for maintaining the sterile environment, and instrument sterile adapters (ISAs) may be mounted in the sheet and positioned between the medical instruments and the robot. The ISAs typically include structures that enable medical instruments to dock with the robot and enable the robot to mechanically actuate docked medical instruments without compromising the sterile environment. A medical instrument may employ latch and release mechanisms to securely dock the medical instrument with a robot via an ISA and to release the medical instrument from the robot and the ISA, for example, when a procedure is complete.

SUMMARY

In accordance with an aspect of the invention, a medical instrument includes a backend with a pair of release levers. The backend may have a docking feature shaped to engage a docking structure such as a sterile adaptor or a docking port of a robot. Each of the release levers may be coupled to the backend and may include an activation feature (e.g., a release button) on one side of the backend and a push tab extending from an opposite side of the backend. Movement of the activation features moves the push tabs from being aligned with the docking feature to pushing on the docking structure, permitting the medical instrument to be disengaged from the docking structure.

In accordance with another disclosed aspect, a medical instrument includes a backend with a release mechanism. The backend may have a docking feature shaped to engage a docking structure. The release mechanism includes a pair of release levers and a pair of push tabs. A first release lever has an activation feature, such as a button, on one side of the backend, and a second release lever has another activation feature on an opposite side of the backend. Similarly, a first

push tab of the push tabs extends from one side of the backend, and a second push tab of the push tabs extends from the other side of the backend. A coupling of the two release levers may limit a movement of the activation features. The movement of the activation features causes the push tabs to move from being aligned with the docking feature to pushing on the docking structure.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A shows an implementation of a medical system employing removable instruments.

FIG. 1B shows a medical instrument being slid into a docking port of a medical system.

FIG. 1C shows the medical instrument in accordance with the implementation of FIG. 1B when docked in a docking port.

FIG. 1D shows the docked medical instrument in accordance with the implementation of FIG. 1C when a user moves or activates a release mechanism.

FIGS. 2A and 2B show perspective and top views of an example implementation of a medical instrument.

FIG. 3A shows a cut away view revealing one implementation of a release mechanism that may be inside a backend of the instrument of FIGS. 2A and 2B.

FIG. 3B shows a perspective view of the release mechanism of FIG. 3A without surrounding components in a backend of an instrument.

FIGS. 4A, 4B, and 4C respectively show front, side, and perspective views of one implementation of a release lever.

FIGS. 5A, 5B, and 5C are cross-sectional views illustrating installation of a release mechanism in a medical instrument.

FIG. 5D shows a cross-sectional view illustrating the effect of a user depressing or activating a release mechanism in accordance with the implementation of FIGS. 5A, 5B, and 5C.

FIG. 6A shows a perspective view of a release mechanism in accordance with an implementation employing multi-piece release levers.

FIG. 6B is a cutaway view illustrating the interconnection of release buttons on opposite sides of a medical instrument using the release mechanism of FIG. 6A.

FIG. 6C is a side view illustrating the connection an release button to a link and a push rod that form an implementation of a multi-piece release lever.

FIG. 6D shows an instrument containing the release mechanism of FIG. 6A when the instrument is engaged with an instrument sterile adapter and a docking port of a robot.

FIG. 6E shows an instrument containing the release mechanism of FIG. 6A when the release mechanism is activated.

The drawings illustrate examples for the purpose of explanation and are not of the invention itself. Use of the same reference symbols in different figures indicates similar or identical items.

DETAILED DESCRIPTION

A release mechanism for a medical instrument may push a sterile adapter retention plate away from the medical instrument using coupled levers to simultaneously press on opposite edges of the retention plate. The release mechanism particularly pushes the retention plate away symmetrically, so that the medical instrument does not get cocked to one side, which could prevent the medical instrument from sliding smoothly out of an instrument sterile adapter (ISA).

The medical instrument may engage the ISA by being slid into a position at which spring action of the ISA or other docking structure may push projections on opposite sides of the medical instrument or the ISA into notches in the other of the ISA and the medical instrument. Disengaging the instrument may thus require the release mechanism to push the projections out of the notches, so that the instrument can be slid out of (or otherwise removed from) the ISA. Release buttons or other activation features of the release mechanism may be positioned on a housing of the medical instrument so that a user grasping the housing when removing the instrument can naturally activate the release mechanism and reliably push the retention plate and the instrument apart using a grasping force within desired limits.

FIG. 1A shows an example of a medical system 100 including replaceable medical instruments 110 that may contain release mechanisms such as disclosed herein. System 100, which may, for example, include a da Vinci® Surgical System commercialized by Intuitive Surgical, Inc., employs multiple surgical instruments 110, each of which is mounted in a docking port 120 on a manipulator arm 130 of a robot 140. A sterile barrier (not shown in FIG. 1A) including a drape and instrument adaptors may be between a patient (not shown) and robot 140. Robot 140, including manipulator arms 130 and docking ports 120, may thus be outside a sterile environment for the patient, while instruments 110 are in the sterile environment on the patient's side of the sterile barrier. Accordingly, robot 140 may not need to be sterilized for medical procedures. In contrast, instruments 110, which directly contact the patient, are compact and removable so that medical instruments 110 may be removed and sterilized or replaced between medical procedures performed using system 100. The sterile barrier may be removed from robot 140 after medical instruments 110 are removed, but the sterile barrier is typically disposed of and replaced between medical procedures.

Instruments 110 may vary in structure and purpose but may still be interchangeable, so that a user can select and mount various instruments 110 in docking ports 120 of robot 140 as needed for a particular medical procedure and can swap instruments 110 during a medical procedure to provide desired clinical functions. Each instrument 110 generally includes an end effector or distal tool 112, an instrument shaft 114, and a backend. Distal tools 112 may have different designs to implement many different functions. For example, distal tools 112 for different instruments 110 may have many different shapes or sizes and may include forceps, graspers, scalpels, scissors, cautery tools, or needle drivers to name a few possibilities. Instruments 110 having different distal tools 112 may be mounted on different arms 130 of robot 140 and may work cooperatively at the same work site. An endoscopic camera, for example, a stereoscopic camera, can also be mounted on an arm to provide visual information, particularly images of the work site in which distal tools 112 of instruments 110 may be operating.

Docking ports 120 of robot 140 may include actuators such as drive motors that provide mechanical power to actuate mechanical structures in instruments 110 via drive couplings that connect the actuators through an ISA to inputs of instruments 110. FIG. 1B illustrates an instrument 110 beginning to engage a docking port 120. An ISA 150, which may be part of a sterile barrier, may be mounted on docking port 120 before the engagement process begins. In particular, ISA 150 may be attached to docking port 120. In the implementation shown, ISA 150 includes an outer frame 158 around a movable plate 157 containing rotatable disks 156. Outer frame 158, which is shown in cross-section to better

illustrate the structure of ISA 150, may be disposed in a sheet or drape portion 160 of the sterile barrier and is sized and shape to fit docking port 120. Inner frame 157, which is sometimes referred to herein as a retention plate, may be movable relative to outer frame 158 to adjust to variability in the position of actuators in docking port 120 so that engagement features 122 of the actuators in docking port 120 may couple to disks 156 in ISA 150. In particular, springs (not shown) in the docking port 120 may provide spring loading that pushes engagement features 122 onto disks 156 and plate 157. Disks 156 in turn engage engagement features on input spindles 220 of instrument 110 when instrument 110 is fully docked.

ISA 150 further includes rails 152 shaped to guide side rails 232 and projections 234 on opposite sides of backend 116 of instrument 110 as backend 116 is being slid into ISA 150. When projections 234 reach respective notches 154 in ISA 150, rails 152 push side rails 232 against retention plate 157 or outer frame 158 of ISA 150 and push each projection 234 into its notch 154 in ISA 150 as shown in FIG. 1C. In this docked position, engagement features 122 on drive motors in docking port 120 fit into top features of sealed disks 156 in ISA 150, and engagement features 228 on input spindles of backend 116 fit into bottom features of sealed disks 156 so that drive motors rotating engagement features 122 rotate disks 156 and input spindles 220 of instrument 110. A computer system, which may be connected to or part of the robot 140 that includes docking port 120 and which may be connected to a user interface device (not shown), may receive user commands from a surgeon or other medical personnel and may execute software that controls drive motors or other actuators in docking ports 120 as needed to rotate engagement features 122, disks 156, and input spindles 220 and thereby actuate instruments 110 according to the user commands.

FIG. 1C shows the projection 234 on one side of backend 116 secured in the notch 154 on one side of ISA 150. Projections 234 and notches 154 act as lock features that prevent instrument 110 from being unintentionally removed from ISA 150. In particular, two projections 234 on opposite sides of backend 116 being pressed into notches 154 on opposite sides of ISA 150 may keep backend 116 from disengaging from docking port 120 and from sliding out of ISA 150. To release instrument 110 from docking port 120 and ISA 150, instrument 110 includes a release mechanism that includes interconnected release levers with push tabs 240 on opposite sides of backend 116. When the release mechanism is not activated, each push tab 240 may be aligned with or recessed relative to a side rail 232 so that backend 116 can slide into place and fully engage with ISA 150 and docking port 120 as shown in FIG. 1C. A user may depress release buttons 250 on opposite sides of backend 116 to activate the release mechanism and cause push tabs 240 to move up past respective rails 232 and to thereby push on ISA 150. Push tabs 240 may particularly push on plate 157, so that plate 157 moves away from backend 116 and disengages backend 116 from notches 154 and/or other engagement features of ISA 150. Push tabs 240 may also push on plate 157, so that plate 157 moves disks 156 away from backend 116 and disengages bottoms of disks 156 from input spindles 220 in backend 116. FIG. 1D shows how the action of push tabs 240 may also keep ISA 150 engaged with docking port 120 while decoupling projections 234 from notches 154 so that medical instrument 110 can be slid out of ISA 150. ISA 150 thus remains on docking port 120 when instrument 110 is released and removed.

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FIGS. 2A and 2B illustrate a more detailed example of an implementation of a medical instrument 110. FIG. 2A particularly shows a perspective view of an implementation having an elongated instrument shaft 114 that extends from backend 116 of instrument 110 and having a distal tool 112 at the distal end of instrument shaft 114. In the illustrated implementation, distal tool 112 has six degrees of freedom of movement relative to backend 116. Specifically, the six degrees of freedom may correspond to: pitch and yaw rotations of a portion of tool 112 about two respective perpendicular axes 201 and 202 associated with a first joint or wrist mechanism 211; pitch and yaw rotations or movement of jaws 213 relative to two respective perpendicular axes 203 and 204 associated with a second joint or wrist mechanism 212; opening or closing movement 205 of jaws 213 for “grip” actuation; and “roll” rotations of instrument shaft 114 about its central length axis 206. Other medical instruments may have more, fewer, or different degrees of freedom of movement.

Backend 116 as shown in FIG. 2B has six input spindles 221 to 226 with engagement features that are arranged in a base plate 230 and are shaped to engage actuators, e.g., engagement features of drive motors, in a docking port of a robot via an intervening ISA as illustrated in FIGS. 1B, 1C, and 1D. In general, each input spindle 221 to 226 may be part of a mechanism for actuation of a corresponding degree of freedom of movement of instrument 110. The assignment of input spindles 221 to 226 to corresponding degrees of freedom must be known to the robot, so that the robot can identify and use the correct actuator or actuators to rotate the input spindle or spindles that exercise desired degree or degrees of freedom of instrument 110, but assignment of input spindles 221 to 226 to particular degrees of freedom of movement can be otherwise defined by an arbitrary standard or convention.

Base plate 230 further includes side rails 232 on opposite sides of base plate 230, and side rails 232 may be positioned and shaped for sliding of backend 116 into an ISA as illustrated in FIG. 1B. Base plate 230 may further include projections 234 that may be contiguous with respective rails 232 or separated from rails 232. Each projection 234 may be positioned and shaped to slip into a corresponding notch in the ISA so that projections 234 when in respective notches prevent backend 116 from being slid out of the ISA. Removal of the instrument may thus require lifting projections 234 out of the notches in the ISA as described above with reference to FIG. 1D.

FIG. 3A shows a cutaway view of the instrument backend 116 of FIGS. 2A and 2B in accordance with an implementation in which a release mechanism employs a pair of identical interconnected release levers 310 that each function as a release element of a release mechanism 300. FIG. 3B shows release mechanism 300 including release levers 310 but without other structures of backend 116 that may obscure the view of the interaction of release levers 310. In release mechanism 300, each release lever 310 has a one-piece construction and may be a monolithic or molded structure that integrates and connects one push tab 240 and one release button 250. Each lever 310 further includes a pair of pivot arms 320 and a lifter arm 330. Each button 250 is coupled to a crossbar extending between a pair of pivot arms 320. The two pivot arms 320 on each release lever 310 have pivots 322 (which can be implemented as axles) that extend along a desired rotation axis of the release lever 310. In backend 116 of FIG. 3A, pivots 322 of a release lever 310 rotatably attach the release lever 310 to base plate 230. Pivot arms 320 and buttons 250 may particularly be positioned

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toward outer edges of backend 116, so that actuation mechanisms such as input spindles 221 to 224 can occupy the central portion of backend 116 as shown in FIG. 3A. The lifter arm 330 of each release lever 310 may extend from the release button 250 of the release lever 310, between the input spindles of backend 116, across backend 116, and upward so that the push tab 240 at the end of lifter arm 330 extends from a side of backend 116 that is opposite the side on which the button 250 resides. Accordingly, a user pushing on a button 250 on one side of backend 116 causes release lever 310 to rotate and the push tab 240 on the opposite side of backend 116 to move.

FIGS. 4A, 4B, and 4C respectively show front, side, and perspective views of an implementation of release lever 310 that can be used in release mechanism 300 of FIG. 3B. Release lever 310 may be a one-piece or monolithic structure and may be molded from a plastic or other sufficiently durable material. The release lever implementation of FIGS. 4A, 4B, and 4C includes guide channel 332 and a guide pin 334 on lifter arm 330 to allow interconnection of a pair of identical release levers 310. That is, to reduce parts and save manufacturing cost, identical release levers are used for each side, and the release levers are configured to work together in this hermaphroditic configuration. In particular, in assembling release mechanism 300 of FIG. 3B, guide pin 334 on the lifter arm 330 of one release lever 310 may be inserted into the guide channel 332 on the lifter arm 330 of the other release lever 310. Guide channel 332 may be arc-shaped to accommodate or guide rotations of both release levers 310 about their respective pivots 322. One of the advantages of the pin-in-slot configuration is that guide channels 332 control the movement of pins 334, and so both release levers 310 are forced to move symmetrically. Thus, both tabs 240 may push upward on the sterile adapter retention features with the same displacement, and the instrument stays level relative to the sterile adapter and does not get cocked to one side when the tabs 240 push upward. Having two linked release levers 310 on opposite sides of the instrument also means that the force applied to one lever 310 is counteracted in the other 310, and user activation of release mechanism 300 does not produce a net force that might otherwise push the instrument left or right.

Lifter arm 330 on a release lever 310 extends from an off-center location on release button 250 to position guide channel 332 and pin 334 for connection to the lifter arm 330 of an identical release lever 310. In some implementations, a spring may be inserted in a cavity formed by adjacent guide channels 332 to push guide pins 334 apart when no external pressure is applied to buttons 250.

A user can operate or activate release mechanism 300 of FIGS. 3A and 3B by grasping a backend 116 and depressing both buttons 250 simultaneously. Depressing buttons 250 causes both release levers 310 to rotate about their individual axes defined by the connection of pivots 322 to base plate 230 and guide pins 334 in channels 332, and the rotations of release levers 310 lift push tabs 240 relative to base plate 230.

The implementation of release lever 310 of FIGS. 4A, 4B, and 4C when assembled into the release mechanism 300 of FIGS. 3A and 3B has several advantageous features. In particular, pivot arms 320 may provide a long radius for rotation of button 250 about pivots 322 so that button 250 moves in a substantially linear direction when depressed. Further, the moment or radius of rotation of push tab 240 about pivot 322 may be roughly equal to moment or radius of rotation of button 250 about pivot 322, so that the force a user applies on button 250 may be about the same as the

force that push tab **240** exerts on the retention plate when release mechanism **300** is activated. Further, the one-piece structure of release levers **310** and the hermaphroditic engagement of release levers **310** in release mechanism **300** allow both release levers **310** to be identical parts, which may reduce the part count and cost of a medical instrument. Release levers **310** can also extend across an instrument backend and wrap around backend mechanical systems such as the input spindles and cables shown in FIG. **3A**. Release mechanism **300** may, therefore, be installed in an instrument backend after all cabling and centrally located mechanical systems are assembled, which simplifies the instrument assembly process because release mechanism **300** does not interfere with assembly of other backend mechanical systems.

FIGS. **5A**, **5B**, and **5C** illustrate a process of assembling a pair of identical release levers **310A** and **310B** in a base plate **230** of an instrument backend to form one implementation of an instrument release mechanism **300**. Each release lever **310A** or **310B** may have a one-piece molded construction including two pivot arms **320A** or **320B** and a lifter arm **330A** or **330B** as described above. Release levers **310A** and **310B** may be engaged with each other before being engaged with base plate **230**. In particular, a guide pin **334A** of release lever **310A** may be inserted in a guide channel **332B** of release lever **310B**, and a guide pin **334B** of release lever **310B** may be inserted in a guide channel **332A** of release lever **310A**. A spring **510** may be installed between guide pins **334A** and **334B** in a cavity formed by guide channels **332A** and **332B** and may improve the look and feel of release buttons **250A** and **250B** in the assembled release mechanism **300**.

Each guide channel **332A** and **332B** may extend beyond the range of motion that guide pins **334B** and **334A** have for normal use of release mechanism **300**. In particular, an assembly process may initially separate pivots **322A** and **322B** by a greater distance than required for normal use. The wider separation allows pivots **322A** and **322B** to be inserted into L-shaped slots **236** in base plate **230** as shown in FIG. **5A**. Pivots **322A** and **322B** may then be slid up into slots **236** as shown in FIG. **5B** before buttons **250A** and **250B**, and pivots **322A** and **322B** shift inward for use as shown in FIG. **5C**. Accordingly, guide channels **332A** and **332B** may have extensions that can accommodate the greater separation initially needed between guide pins **334B** and **334A** during assembly of release mechanism **300**.

FIG. **5C** also illustrates a housing **520** for the instrument backend that encloses the mechanics of the backend including release mechanism **300**. Housing **520** may include tabs **522** that fit into the L-shaped slots in base plate **230** and help to capture the release lever pivots **322A** and **322B** in their working positions. Housing **520** also captures release buttons **250A** and **250B** at the proper separation so that push tabs **240A** and **240B** are substantially aligned with rails **232**. The bottom cutaways in FIG. **5C** show how spring **510** compresses to fit between guide pins **334A** and **334B** in the assembled position, which loads spring **510** to push the release buttons **250A** and **250B** apart. FIG. **5D** illustrates how pushing release buttons **250A** and **250B** inward rotates release levers **310A** and **310B** about respective pivots **322A** and **322B**, raising push tabs **240A** and **240B** above rails **232** so that push tabs **240A** and **240B** can push on an ISA or other docking structure.

FIG. **6A** shows a release mechanism **600** in accordance with an implementation that positions push tab **240A** on the same side as release button **250A** on an instrument backend, and likewise push tab **240B** is on the same side as release

button **250B** on the instrument backend. In release mechanism **600**, release buttons **250A** and **250B** are interconnected using alignment features that limit buttons **250A** and **250B** to moving linearly toward or away from each other. For example, FIG. **6B** shows an implementation in which slides **652A** and **652B** on release buttons **250A** and **250B** fit into slots **654B** and **654A** in the opposite release buttons **250B** and **250A**, so that buttons **250A** and **250B** can only move relative to each other along a length axis of slides **652A** and **652B**. As described above, the release mechanism assemblies for each side are identical and are configured to work together.

Each release button **250A** or **250B** in release mechanism **600** of FIG. **6A** further couples to a three-pivot link **610A** or **610B**. FIG. **6C**, for example, shows a release button **650**, which may be identical to button **250A** or **250B**, having a slot and pin coupling **612** to a three-pivot link **610**, which may be identical to link **610A** or **610B** of FIG. **6A**. A second pivot **614** of link **610** may be fixed on a chassis of a backend of an instrument containing the release mechanism **600**, and a third pivot **616** of link **610** couples link **610** to a push rod **620**. During use of release mechanism **600**, pushing on release button **650** rotates link **610** and thereby pushes push rod **620** upward. Accordingly, in release mechanism **600** of FIG. **6A**, pushing on button **250A** or **250B** rotates the connected link **610A** or **610B** and moves push rod **620A** or **620B** in a generally upward direction. Push tabs **240A** and **240B** may have guide features that restrict the movement of tabs **240A** and **240B** to desired directions, e.g., up and down. Thus as shown, each assembly of button **250**, slides **652**, link **610**, and push rod **620** functions as a release element of the release assembly.

FIG. **6D** shows an implementation of an instrument backend **116** containing release mechanism **600** of FIG. **6A** when instrument backend **116** is engaged with a docking port **120** via an ISA **150**. In the engaged configuration as shown in FIG. **6D**, push tabs **240** may be aligned with side rails **232** on the housing of backend **116**. As described above with reference to FIG. **1C**, backend **116** and ISA **150** may have complementary projections and notches that fit together and prevent backend **116** from being slid out of ISA **150** while the projections and notches are engaged. When a user presses release both buttons **250A** and **250B** as shown in FIG. **6E**, resulting rotations of links **610A** and **610B** move push rods **620A** and **620B** upward, so that push tabs **240A** and **240B** push on ISA **150**, which can disengage projections and notches on backend **116** and ISA **150**. Pushing release buttons **250A** and **250B** may thus release backend **116** from ISA **150**, while ISA **150** and docking port **120** remain engaged.

Although particular implementations have been disclosed, these implementations are only examples and should not be taken as limitations. Various adaptations and combinations of features of the implementations disclosed are within the scope of the following claims.

What is claimed is:

1. A release mechanism for a medical instrument, the release mechanism comprising:
 - an activation feature positioned on a first side of a backend of the medical instrument;
 - a push tab extending from a second side of the backend, the second side being different than the first side; and
 - a lever coupling the activation feature and the push tab, the lever including a pivot arm and a lifter arm, wherein the push tab is configured to push on a portion of a docking structure in response to a movement of the

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activation feature to separate the portion of the docking structure and the backend of the medical instrument.

2. The release mechanism of claim 1, wherein: the docking structure is a retention plate of an instrument sterile adapter; 5

the instrument sterile adapter is a portion of a sterile barrier; and

the sterile barrier separates the medical instrument from a manipulator docking port. 10

3. The release mechanism of claim 1, wherein: the activation feature, the push tab, the pivot arm, and the lifter arm are formed as a monolithic structure.

4. The release mechanism of claim 1, wherein: the pivot arm extends from the activation feature to a pivot; and 15

the pivot is configured to be movably coupled to a base plate of the backend on the first side.

5. The release mechanism of claim 4, wherein: a radius of rotation of the push tab about the pivot is substantially equal to a radius of rotation of the activation feature about the pivot. 20

6. The release mechanism of claim 1, wherein: the lifter arm extends from the activation feature to the second side of the backend; and 25

the push tab extends from the lifter arm in a position to exert a force on the portion of the docking structure.

7. The release mechanism of claim 1, wherein the activation feature is a first activation feature, the push tab is a first push tab, the lever is a first lever, the release mechanism further comprising: 30

a second activation feature positioned on the second side of the backend;

a second push tab extending from the first side of the backend; and 35

a second lever coupling the second activation feature and the second push tab,

wherein the first push tab and the second push tab are configured to push on the portion of the docking structure in response to the movement of the first activation feature and a movement of the second activation feature to separate the portion of the docking structure and the backend of the medical instrument. 40

8. The release mechanism of claim 7, wherein: the first lever includes a first pin and a first guide channel; 45

the second lever includes a second pin and a second guide channel;

the first pin is positioned within the second guide channel; the second pin is positioned within the first guide channel; and 50

a movement of the first pin is controlled by the second guide channel and a movement of the second pin is controlled by the first guide channel so that the first lever and the second lever move symmetrically in response to the movement of the first activation feature and the second activation feature. 55

9. The release mechanism of claim 8, wherein: each of the first guide channel and the second guide channel have an arc-shape configured to guide a rotation of each of the first lever and the second lever. 60

10. The release mechanism of claim 7, wherein: the first lever is operably coupled to the second lever via a plurality of pins and corresponding guide channels; the operable coupling of the first lever and the second lever causes a reactive force in response to an activation force applied to one of the first lever or the second lever; and 65

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the reactive force counteracts the activation force so that a net force does not push the backend perpendicular to a direction in which the first push tab and the second push tab move.

11. A medical instrument comprising: a backend including a docking feature shaped to engage a retention plate of an instrument sterile adapter; a first release element coupled to the backend and including a first activation feature on a first side of the backend and a first push tab extending from the first side; and a second release element coupled to the backend and including a second activation feature on a second side of the backend and a second push tab extending from the second side of the backend, the second side being opposite to the first side, wherein a movement of the first and second activation features moves the first and second push tabs from being aligned with the docking feature to pushing on the retention plate to separate the retention plate and the backend.

12. The medical instrument of claim 11, wherein: the instrument sterile adapter is a portion of a sterile barrier; and the sterile barrier separates the medical instrument from a manipulator docking port of a surgical system.

13. The medical instrument of claim 11, wherein: the docking feature includes a rail and a lock feature; the rail is shaped to slidably engage with the retention plate; and the lock feature is shaped to engage the retention plate and prevent the medical instrument from being slid relative to the retention plate when engaged.

14. The medical instrument of claim 13, wherein: the lock feature is positioned to be disengaged from the retention plate on a condition that the first and second push tabs are pushed against the retention plate in response to the movement of the first and second activation features.

15. The medical instrument of claim 11, wherein: the first activation feature is a first button; the second activation feature is a second button; and each of the first button and the second button are positioned so that a user can simultaneously depress the first button and the second button in a singular motion by grasping the backend.

16. The medical instrument of claim 11, wherein: the first release element includes a first link and a first pushrod; the first link includes a first pivot coupled to the backend and a second pivot coupled to the first activation feature; the first link rotates about the first pivot in response to the movement of the first activation feature; the first pushrod is coupled to a third pivot of the first link; and the first push tab extends from the first pushrod.

17. The medical instrument of claim 16, wherein: the second release element includes a second link and a second pushrod; the second link includes a fourth pivot coupled to the backend and fifth pivot coupled to the second activation feature; the second link rotates about the fourth pivot in response to the movement of the second activation feature; the second pushrod is coupled to a sixth pivot of the second link; and the second push tab extends from the second pushrod.

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18. The medical instrument of claim **11**, wherein:
the first release element and the second release element
are operably coupled together; and
the first activation feature and the second activation
feature are limited to moving linearly toward or away 5
from each other.

19. The medical instrument of claim **11**, wherein:
the first release element includes a first slide member and
defines a first slot;
the second release element includes a second slide mem- 10
ber and defines a second slot;
the first slot is sized to receive the second slide member;
and
the second slot is sized to receive the first slide member.

20. The medical instrument of claim **11**, wherein: 15
the first push tab includes a first guide feature;
the first guide feature restricts a movement of the first
push tab to a direction that is orthogonal to a direction
of movement of the first activation feature and the
second activation feature; 20
the second push tab includes a second guide feature; and
the second guide feature restricts a movement of the
second push tab to the direction that is orthogonal to the
direction of movement of the first activation feature and
the second activation feature. 25

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